CMPG 325 COMPUTER NETWORKS INDIVIDUAL PROJECT

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1. INTRODUCTION

This report documents the analysis, design, implementation, testing and verification of a network for Kabelo Telecom Installations (Rustenburg), a telecommunications company. The network was built and simulated in Cisco Packet Tracer using the 10.55.0.0/16 address block allocated to this project. The client had no proper network design for its new floor or area, and guest users risked reaching internal resources. The solution is a hierarchical, VLAN-segmented network with a Layer 3 core, floor distribution and access switches, an edge router, DHCP, inter-VLAN routing, an access control list that isolates guests, and Port Security on user-facing switch ports. The project was delivered in stages. Milestone 1 covered requirements, topology, IP plan and the initial GitHub repository. Milestone 2 covered the implementation and testing. This report consolidates both into one document and explains what was configured, why it was configured, and how each part was verified.

2. CLIENT REQUIREMENTS

2.1. The client

ITEM	DETAIL
Organisation	Kabelo Telecom Installations
Industry	Telecommunications
Users	Staff, Guest, Servers and Management
Tool	Cisco Packet Tracer (simulation, so no hardware cost)

2.2. The Problem

The client has no proper network design for its new floor or area. Guest users could potentially access internal resources, and unauthorized devices could be plugged into available switch ports. The goal is reliability, secure connectivity with proper segmentation and coverage expansion.

2.3. Requirements

REQUIREMENT	HOW IT IS ADDRESSED
Users: staff, guests, servers	Separate VLANs for each group
Security: guest isolation, port security, VLAN segmentation	VLANs, GUEST_ISOLATION ACL, Port Security with sticky MAC

3. NETWORK DESIGN

3.1. Design approach

The network follows the three-layer hierarchical model, which keeps functions separate and makes growth simple.

- **Core layer:** one Cisco 3650-24PS Layer 3 switch (Core-3650). It performs inter-VLAN routing, DHCP and ACL filtering.
- **Distribution layer:** four Cisco 3560-24PS switches, one per floor (Floor 1-Dist to Floor 4-Dist).
- **Access layer:** four Cisco 2960-24TT switches, one per floor, connecting end devices.
- **Edge:** one Cisco ISR4331 router connected to the core for WAN/internet-facing connectivity.

The physical topology is an extended star: the core sits at the centre, each distribution switch is a branch, and each access switch hangs below its distribution switch. This was chosen because a fault on one floor does not affect the others, and a new floor is added by attaching another branch to the core without redesigning anything.

3.2. Physical topology

Device	Model	Quantity	Role
Core switch	3650-24PS	1	Layer 3 routing, DHCP, ACL
Distribution switches	3560-24PS	4	One per floor
Access switches	2960-24TT	4	One per floor
Router	ISR4331	1	Edge router
Staff/management PCs	PC-PT	19	Floors 1-3 staff, Floor 4 management/new floor
Guest PCs	PC-PT	3	Floors 1-3
Servers	Server-PT	4	One per floor
Wireless access points	Access Point-PT	5	One per floor, two on Floor 4

All links use copper straight-through cable: core to distribution, distribution to access, and access to PCs, servers and access points.

3.3. Logical topology

Logically, the network is VLAN-segmented. Although the cabling is a star, traffic is separated into broadcast domains by VLAN, so a Staff PC on Floor 1 and a Staff PC on Floor 3 are in the same VLAN even though they sit on different branches. Devices in different VLANs can only communicate by being routed through the core, where policy such as the guest ACL is applied.

- **Trunk links:** every uplink (core to distribution, distribution to access) is an 802.1Q trunk that carries all four VLANs over a single cable.
- **Access links:** ports to PCs and servers belong to exactly one VLAN.
- **Inter-VLAN routing:** performed on the core using one Switch Virtual Interface (SVI) per VLAN.
- **DHCP:** four pools on the core, one per VLAN

3.4. IP addressing and VLAN plan

The 10.55.0.0/16 block was divided into /24 subnets, one per VLAN, giving 254 usable hosts per VLAN, which leaves room to grow.

VLAN ID	Name	Subnet	Usable range	Gateway (SVI on core)	Purpose
10	STAFF	10.55.1.0/24	10.55.1.1 – 10.55.1.254	10.55.1.1	Staff PCs, Floors 1-3
20	GUEST	10.55.2.0/24	10.55.2.1 – 10.55.2.254	10.55.2.1	Guest PCs and guest Wi-Fi
30	SERVERS	10.55.3.0/24	10.55.3.1 – 10.55.3.254	10.55.3.1	All four servers
40	MANAGEMENT	10.55.4.0/24	10.55.4.1 – 10.55.4.254	10.55.4.1	Floor 4, the new floor added for CR2

The router link uses a dedicated point-to-point subnet, **10.55.254.0/30**: Core-3650 Gi1/0/1 = 10.55.254.1 and Router Gi0/0/0 = 10.55.254.2.

Reserved addresses (excluded from DHCP): the four gateway addresses (10.55.1.1, 10.55.2.1, 10.55.3.1, 10.55.4.1) and 10.55.3.1 – 10.55.3.10 for servers.

3.5. Changes from the Milestone 1 plan

The Milestone 1 plan labelled the VLANs 1-4 with the fourth subnet as a New Floor Staff VLAN. During implementation the VLANs were numbered 10, 20, 30 and 40 (avoiding VLAN 1, which is the default VLAN) and the fourth VLAN was named MANAGEMENT. The subnets and gateways stayed exactly as planned, and VLAN 40 serves the Floor 4 area added under CR2.

4. IMPLEMENTATION

All configuration was done on the switch and router command-line interfaces in Packet Tracer and saved with copy running-config startup-config.

4.1. VLAN creation

VLANs 10 (STAFF), 20 (GUEST), 30 (SERVERS) and 40 (MANAGEMENT) were created on the core, all four distribution switches and all four access switches so the same VLANs exist everywhere.

```
VLAN 10
 name STAFF
VLAN 20
 name GUEST
VLAN 30
 name SERVERS
VLAN 40
 name MANAGEMENT
```

Verified with show VLAN brief on each switch (Milestone 2, Figures 2-7).

4.2. Access port assignment

On each access switch, user-facing ports were set to access mode and assigned to a single VLAN.

Floor	Staff / user PC ports	VLAN	Server port	VLAN
1, 2, 3	Fa0/1, Fa0/4 – Fa0/7	10 (STAFF)	Fa0/2	30 (SERVERS)
4	Fa0/1, Fa0/4 – Fa0/6	40 (MANAGEMENT)	Fa0/2	30 (SERVERS)

```
interface range fastEthernet 0/1, fastEthernet 0/4-7
 switchport mode access
 switchport access VLAN 10
interface fastEthernet 0/2
 switchport mode access
 switchport access VLAN 30
```

Guest PCs on Floors 1-3 sit on dedicated VLAN 20 access ports. Verified in Milestone 2, Figures 8-11.

4.3. Trunk links

The core's four distribution-facing ports (Gi1/0/2 to Gi1/0/5) were configured as 802.1Q trunks allowing VLANs 10, 20, 30 and 40. The distribution uplinks and the access switch uplinks (Gi0/1) were trunked the same way. The access point ports were also trunked to carry the Staff and Guest VLANs (VLANs 10 and 20).

```
interface range gigabitEthernet 1/0/2-5
 switchport mode trunk
 switchport trunk allowed VLAN 10,20,30,40
```

Verified with show interface trunk (Milestone 2, Figures 12-20).

4.4. Inter-VLAN routing

IP routing was enabled on Core-3650 and one SVI created per VLAN. Each SVI is the default gateway for its VLAN.

```
IP routing
interface VLAN 10
IP address 10.55.1.1 255.255.255.0
 no shutdown
(repeated for VLAN 20 = 10.55.2.1, VLAN 30 = 10.55.3.1, VLAN 40 = 10.55.4.1)
```

show IP interface brief shows all four SVIs up/up, and show IP route shows four connected /24 routes (Milestone 2, Figures 21-23).

4.5. DHCP

DHCP was configured on the core. Gateways and the server range were excluded, and one pool was created per VLAN with the correct network and default router.

```
IP DHCP excluded-address 10.55.1.1
IP DHCP excluded-address 10.55.2.1
IP DHCP excluded-address 10.55.3.1 10.55.3.10
IP DHCP excluded-address 10.55.4.1
IP DHCP pool STAFF
 network 10.55.1.0 255.255.255.0
 default-router 10.55.1.1
(similar pools for GUEST, SERVERS and MANAGEMENT)
```

Because clients are on different subnets from the core, the SVI acts as the DHCP relay point, so each VLAN's broadcast is answered by its own pool. All four floors received correct addresses, masks and gateways (Milestone 2, Figures 28-36). The binding table shows Staff PCs on Floors 1-3 leasing 10.55.1.2 – 10.55.1.16 and Floor 4 PCs leasing 10.55.4.2 – 10.55.4.5.

4.6. Edge router and static routing

The core port Gi1/0/1 was converted to a routed port (no switchport) with 10.55.254.1/30, and the router's Gi0/0/0 was given 10.55.254.2/30. Because the router does not learn the internal VLANs automatically, static routes were added for all four subnets via the core.

```
Router(config)# IP route 10.55.1.0 255.255.255.0 10.55.254.1
Router(config)# IP route 10.55.2.0 255.255.255.0 10.55.254.1
Router(config)# IP route 10.55.3.0 255.255.255.0 10.55.254.1
Router(config)# IP route 10.55.4.0 255.255.255.0 10.55.254.1
```

5. DESIGN CONSTRAINT

Constraint: guest users must not reach internal resources.

Three layers work together:

1. **VLAN separation:** guests are in VLAN 20 on their own subnet, so they are in a different broadcast domain from Staff, Servers and Management.
2. **Routing is the only path between VLANs,** and it happens on the core, where policy can be enforced.
3. **Extended ACL GUEST_ISOLATION** applied inbound on the Guest SVI (interface VLAN 20), blocks guest traffic to the internal subnets.

```
IP access-list extended GUEST_ISOLATION
 deny IP 10.55.2.0 0.0.0.255 10.55.1.0 0.0.0.255
 deny IP 10.55.2.0 0.0.0.255 10.55.3.0 0.0.0.255
 deny IP 10.55.2.0 0.0.0.255 10.55.4.0 0.0.0.255
 permit IP 10.55.2.0 0.0.0.255 any
interface VLAN 20
 IP access-group GUEST_ISOLATION in
```

The three deny statements come first because ACLs are processed top-down and stop at the first match. The final permit statement lets guests reach everything else, such as external or internet-bound traffic, since an ACL otherwise ends with an implicit deny.

Verification: A Guest PC pinging a Staff PC (10.55.1.x), a Server (10.55.3.x) and a Management PC (10.55.4.x) received Destination host unreachable from the gateway 10.55.2.1 with 100% loss. show access-lists shows match counters increasing on the deny lines, proving the ACL is doing the blocking (Milestone 2, Guest Isolation figures).

6. ASSIGNED NETWORK CHALLENGE

6.1. What Port Security is

Port Security is a Layer 2 switch feature that limits which devices may use a switch port, based on their MAC addresses. If a device that is not allowed connects, the switch takes a configured action.

6.2. Why it is appropriate

The access switches connect directly to end-user devices. Without Port Security, anyone could plug an unauthorized laptop into a spare wall port and join the Staff or Management VLAN. This directly supports the client's goal of preventing unauthorized device access. Limiting each user port to one MAC address means one port equals one authorized device. Sticky learning makes this practical because the switch learns the legitimate device's MAC address automatically rather than an administrator typing in every MAC address.

6.3. What was configured

Port Security was applied for the PC-facing access ports of all four access switches.

Setting	Value	Reason
Mode	switchport mode access	Port Security only works on static access (or trunk) ports, not dynamic ports
Maximum MAC addresses	1	One authorized device per port
MAC learning	Sticky	Learns and remembers the first device seen
Violation mode	Restrict	Drops unauthorised frames, logs the event and counts violations without shutting the port

```
interface range fastEthernet 0/1, fastEthernet 0/4-7
 switchport mode access
 switchport port-security
 switchport port-security maximum 1
 switchport port-security mac-address sticky
 switchport port-security violation restrict
```

Floors 1, 2 and 3 secure Fa0/1 and Fa0/4 – Fa0/7. Floor 4 secures Fa0/1 and Fa0/4 – Fa0/6 (four PCs). Trunk ports, server ports and uplinks were deliberately not secured, because trunks legitimately carry many devices' traffic.

6.4. Violation modes considered

Mode	Behaviour	Notes
Protect	Drops violating frames silently	No logging or counter, so less visibility

Mode	Behaviour	Notes
Restrict (chosen)	Drops frames, sends a log message and increments the violation counter	Keeps the legitimate user online and gives the administrator evidence
Shutdown	Puts the port in err-disabled state	Strongest, but one stray device can take a port down and needs manual recovery

Restrict was chosen because it balances security with availability and gives visible evidence of attempts.

6.5. Verification

- **After configuration,** show port-security listed all secured ports with Maximum = 1, Violation = Restrict and 0 current addresses (Milestone 2, Figure 37).
- **After generating traffic,** each PC pinged its gateway, so the switch learned its MAC. show port-security address then listed each MAC as **Secure Sticky**, one per port, in the correct VLAN (VLAN 10 on Floors 1-3, VLAN 40 on Floor 4).
- **Interface detail:** show port-security interface fa0/1 on Floor 4 reported Port Security Enabled, Status Secure-up, Mode Restrict, Maximum 1, Total MAC 1, Sticky MAC 1 and the learned address 0001.6395.D73C in VLAN 40.
- **Violation test:** an unauthorized device was connected to a secured port; its traffic was dropped and the violation counter increased (see the video demonstration and the evidence in the GitHub portfolio).

6.6. Limitations

Sticky MACs held only in the running configuration are lost on reload unless saved. The configuration was saved to startup-config. MAC addresses can be spoofed, so Port Security complements, but do not replace measures such as 802.1X authentication.

7. CHANGE REQUEST

CR2: the client takes over an additional floor or area of the building and needs coverage there.

The design accommodates this change without redesigning the existing network. Floor 4 is an additional area, and it was added as a new branch of the extended star:

- A new Floor 4-Dist (3560-24PS) switch connects to the core by a trunk.
- A new Floor 4 access switch (2960-24TT) connects below it with a trunk.
- Floor 4 has its own VLAN (VLAN 40, 10.55.4.0/24) with an SVI gateway at 10.55.4.1 and its own DHCP pool.
- Floor 4 PCs, a server (VLAN 30) and wireless access points provide coverage across the new area.
- The same Port Security configuration was applied to the Floor 4 PC ports.

Floor 4 PCs received DHCP addresses from 10.55.4.0/24 (Milestone 2, Figures 34-35) and can reach the other VLANs through the core (Milestone 2, Figure 25 tests Staff to Management). Nothing on Floors 1-3 had to be changed, which demonstrates the scalability of the hierarchical design

8. TESTING AND VERIFICATION

#	Test	Method	Result
1	VLANs exist on all switches	show VLAN brief	VLANs 10, 20, 30, 40 are active everywhere
2	Trunks operational	show interface trunk	802.1Q trunking on all uplinks
3	SVIs up	show IP interface brief	Vlan10/20/30/40 up/up
4	Routing table	show IP route on core	Four connected /24 networks
5	DHCP on all floors	PC IP configuration, show IP DHCP binding	Correct subnet, mask and gateway on each floor
6	Staff to gateway and Staff to Staff	ping	0% loss
7	Staff to Server (inter-VLAN)	ping	Success (first packet may time out while ARP resolves)
8	Staff to Management (inter-VLAN)	ping	Success
9	Router to all four gateways	ping from router	100% success
10	Guest to Staff / Server / Management	ping from Guest PC	Blocked, Destination host is unreachable, 100% loss
11	ACL hit counters	show access-lists	Deny and permit counters increment
12	Port Security configured	show port-security	Max 1, sticky, restrict on all PC ports
13	Sticky MACs learned	show port-security address	One Secure Sticky MAC per secured port
14	Port Security violation	Connect unauthorized device	Traffic drop, violation counter increments
15	CR2 coverage	Floor 4 DHCP and ping	Floor 4 reachable and working

9. LIMITATIONS AND VERIFICATION

- **DNS:** DHCP clients currently show a DNS server of 0.0.0.0. A DNS server entry should be added to each DHCP pool (pointing to a server in VLAN 30) and the DNS service enabled on the server.
- **Internet access:** the router is connected and reachable from all VLANs, but no ISP link, default route or NAT were simulated.
- **File sharing and VoIP:** the servers are in place and reachable. Dedicated file-sharing and VoIP services (for example a voice VLAN with telephony service on the router) were not part of the tested scope.
- **Security hardening:** use dot1q encapsulation explicitly on all 3560 trunks, change the native VLAN, shut down unused ports, add SSH-secured management, and consider shutdown mode for high-security ports.
- **Redundancy:** a second core or redundant uplinks would remove the single point of failure at the core.

10. Reflection

This project showed how the separate networking topics work together: VLANs create the segments, trunks carry them, the Layer 3 core connects them, DHCP serves them, and ACLs and Port Security protect them. The hierarchical design made troubleshooting straightforward, because I could test layer by layer, starting at the access port, then the trunk, then the SVI. The most valuable lesson was verification discipline: every feature was proven with a show command and a connectivity test, not just configured. Port Security also taught me that a feature can look correct yet show nothing until traffic flows, as seen with the sticky MAC learning. If I did the project again, I would configure the DNS service and a default route earlier and automate configuration with saved text templates to avoid repeating commands on each floor.

11. Conclusion

The network for Kabelo Telecom Installations was designed and implemented as a hierarchical, VLAN-segmented network using 10.55.0.0/16. Staff, Guest, Server and Management traffic are separated into VLANs, carried over 802.1Q trunks, routed by the Layer 3 core and addressed automatically by DHCP. Guest users are isolated from internal resources by the GUEST_ISOLATION ACL, and Port Security with sticky MAC learning and restrict violation mode protects user-facing access ports against unauthorized devices. The additional Floor 4 required by change request CR2 was added as a new branch without changing the existing floors. Testing confirmed connectivity between authorized devices, proper blocking of guest traffic and correct Port Security operation.

Appendix A: Configuration Summary

Device	Key configuration
Core-3650	VLANs 10/20/30/40; trunks on Gi1/0/2-5; SVIs 10.55.x.1; IP routing; four DHCP pools; GUEST_ISOLATION ACL on Vlan20; routed port Gi1/0/1 = 10.55.254.1/30
Floor 1-4 Dist(3560)	VLANs 10/20/30/40; Gi0/1 trunk to core, Gi0/2 trunk to access
Floor 1-3 Access (2960)	Access ports Fa0/1, Fa0/4-7 in VLAN 10, Fa0/2 in VLAN 30, Gi0/1 trunk, Port Security on PC ports
Floor 4 Access (2960)	Access ports Fa0/1, Fa0/4-6 in VLAN 40, Fa0/2 in VLAN 30, Gi0/1 trunk, Port Security on PC ports
ISR4331 Router	Gi0/0/0 = 10.55.254.2/30; four static routes to 10.55.1.0-4.0/24 via 10.55.254.1

Appendix B: Figure Index (from Milestone 2)

Figures	Content
1	Final topology in Packet Tracer
2-7	VLAN creation and verification on core, distribution and access switches
8-11	Access port VLAN assignment, Floors 1-4
12-20	Trunk configuration and verification
21-23	SVI gateways, interface status and routing table
24-27	Inter-VLAN connectivity tests
28-36	DHCP pools, client addresses and binding table
37 onward	Port Security configuration, sticky MAC tables and interface status
Final pages	Router configuration, static routes, Guest VLAN ACL and isolation tests